Optimal Reconfiguration of Systems in Aeronautics

Lecture 12: "Admissible Model Matching"

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Outline

1. Admissible Model Matching
2. B747 Application Example

Outline

1. Admissible Model Matching
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Preliminaries I

Admissible Model Matching (AMM) aims at giving the post-fault closed-loop system $\hat{M}_f$ the property

$$A_f - B_f K_f \in \hat{\mathscr{A}}_f \supseteq \hat{\mathscr{A}}_n \,, \tag{1}$$

where $\hat{\mathscr{A}}_f$ is the problem specification, i.e., a set of post-fault closed-loop matrices such that any solution of

$$\dot{x}(t) = \hat{A}_f x(t) \,, \quad \text{with} \quad \hat{A}_f \in \hat{\mathscr{A}}_f \,, \tag{2}$$

is admissible in a sense that depends on the problem, namely stability and guaranteed performance under given constraints provided the impaired operation, while $\hat{\mathscr{A}}_n$ is a set of non-faulty admissible closed-loop matrices; the difference $\hat{\mathscr{A}}_f \setminus \hat{\mathscr{A}}_n$ is the set of degraded behaviors that would not be accepted in the nominal operation but possibly in the faulty situation.

Preliminaries on Pareto-Optimality I

This introduction provides some results on Pareto-optimality. It is used when the multi-criteria optimization problem for the Pareto-optimal control design is addressed.

It introduces preliminary definitions from the vector-optimization area, while it also gives the definitions of Pareto-optimal points and controls, respectively.

Vector Optimization I

Definition (Convex Set)

A set $\mathscr{C}$ is convex if the line segment between any two points in $\mathscr{C}$ lies in $\mathscr{C}$, i.e., if, for any $\xi_1, \xi_2 \in \mathscr{C}$ and any c with $0 \leq c \leq 1$, there stands

$$c\xi_1 + (1-c)\xi_2 \in \mathscr{C}\,.$$

□

Vector Optimization II

Definition (Convex and Proper Cones)

A set $\mathscr{K}$ is called a cone if, for every $\xi \in \mathscr{K}$ and $c \geq 0$, one has

$$c\xi \in \mathscr{K} .$$

A set $\mathscr{K}$ is a convex cone if it is convex and a cone, which means that, for any $\xi_1, \xi_2 \in \mathscr{K}$ and $c_1, c_2 \geq 0$, there stands

$$c_1\xi_1 + c_2\xi_2 \in \mathscr{K} .$$

A cone $\mathscr{K}$ is called a proper cone if it is convex, closed, namely it contains its boundary, solid, precisely it has nonempty interior, and it is pointed, that is it contains no line, or, equivalently,

$$\xi \in \mathscr{K}, -\xi \in \mathscr{K} \Rightarrow \xi = 0 .$$

□

Vector Optimization III

Remark (On Generalized Inequalities)

A proper cone $\mathscr{K}$ can be used to define a generalized inequality, which is a partial ordering on $\mathbb{R}^n$ defined by

$$\xi_1 \preceq_{\mathscr{K}} \xi_2 \Leftrightarrow \xi_2 - \xi_1 \in \mathscr{K} \,,$$

and write

$$\xi_1 \succeq_{\mathscr{K}} \xi_2 \,, \text{ for } \xi_2 \preceq_{\mathscr{K}} \xi_1 \,.$$

Similarly, an associated strict partial ordering may be defined by

$$\xi_1 \prec_{\mathscr{K}} \xi_2 \Leftrightarrow \xi_2 - \xi_1 \in \operatorname{int} \mathscr{K} \,,$$

and write

$$\xi_1 \prec_{\mathscr{K}} \xi_2 \,, \text{ for } \xi_2 \succ_{\mathscr{K}} \xi_1 \,,$$

where $\operatorname{int}(.)$ represents the interior of the proper cone (the cone without its closure). $\square$

Vector Optimization IV

Definition (Minimum Element)

Given the convex set $\mathscr{C}$, $\xi \in \mathscr{C}$ is the minimum element of $\mathscr{C}$ w.r.t. the generalized inequality $\preceq_{\mathscr{K}}$ if, for every $\tilde{\xi} \in \mathscr{C}$, $\xi \preceq_{\mathscr{K}} \tilde{\xi}$.

Therefore, a point $\xi \in \mathscr{C}$ is the minimum element of $\mathscr{C}$ if and only if

$$\mathscr{C} \subseteq \xi + \mathscr{K} \,,$$

that is $\xi + \mathscr{K}$ denotes all the points that are comparable to ξ and greater than or equal to ξ according to $\preceq_{\mathscr{K}}$; if a set has a minimum element, then it is unique. $\square$

Vector Optimization V

Definition (Minimal Elements)

Given the convex set $\mathscr{C}$, $\xi_1 \in \mathscr{C}$ is a minimal element of $\mathscr{C}$ w.r.t. the generalized inequality $\preceq_{\mathscr{K}}$ if, for $\xi_2 \in \mathscr{C}$, $\xi_2 \preceq_{\mathscr{K}} \xi_1$ only if $\xi_2 = \xi_1$.

Therefore, a point $\xi \in \mathscr{C}$ is a minimal element if and only if

$$(\xi - \mathscr{K}) \cap \mathscr{C} = \{\xi\},$$

that is $\xi - \mathscr{K}$ denotes all the points that are comparable to ξ and less than or equal to ξ according to $\preceq_{\mathscr{K}}$, whereas the only point in common with $\mathscr{C}$ is ξ; a set can have several different minimal elements. $\square$

Pareto optimality I

Remark (On Cone Domination)

Let $\Upsilon \subset \mathbb{R}^{n \times n}$ and $\Gamma \subset \mathbb{R}^{\bar{n}}$ be the decision and objective spaces, respectively, and let $\bar{\Psi} : \Upsilon \to \Gamma$ be an ordinal preference function that maps the decision space into the objective one, which means the vector function that associates an outcome $\bar{\Psi}(\xi_1) \in \Gamma$ with any decision $\xi_1 \in \Upsilon$.

Then decision ξ_1 dominates decision ξ_2 $(\xi_1 \succeq \xi_2)$ if and only if $\bar{\Psi}(\xi_1) \leq \bar{\Psi}(\xi_2)$, which means that $\psi_l(\xi_1) \leq \psi_l(\xi_2)$, for all components $l = 1, 2, \ldots, \bar{n}$; decision ξ_1 strictly dominates decision ξ_2 $(\xi_1 \succ \xi_2)$ if and only if $\bar{\Psi}(\xi_1) \leq \bar{\Psi}(\xi_2)$ and $\bar{\Psi}(\xi_1) \neq \bar{\Psi}(\xi_2)$.

Moreover, given the cone $\mathscr{K}$, an outcome $\bar{\Psi}(\xi_1)$ dominates or is preferred to an outcome $\bar{\Psi}(\xi_2)$ w.r.t. $\mathscr{K}$, $\bar{\Psi}(\xi_1) \succeq_{\mathscr{K}} \bar{\Psi}(\xi_2)$, if and only if $\bar{\Psi}(\xi_2) - \bar{\Psi}(\xi_1) \in \mathscr{K} \setminus \{0\}$, or, equivalently, there exists a direction $\kappa \in \mathscr{K}$, $\kappa \neq 0$, such that $\bar{\Psi}(\xi_2) = \bar{\Psi}(\xi_1) + \kappa$. Then $\bar{\Psi}(\xi_1) \succ_{\mathscr{K}} \bar{\Psi}(\xi_2)$ if $\bar{\Psi}(\xi_1) \succ_{\mathscr{K}} \bar{\Psi}(\xi_2)$ or $\bar{\Psi}(\xi_1) = \bar{\Psi}(\xi_2)$. $\square$

Pareto optimality II

Definition (Optimal Point)

A point ξ^* is optimal if and only if it is feasible and

$$\Gamma \subseteq \bar{\Psi}(\xi^*) + \mathscr{K} \,, \tag{3}$$

that is the set $\bar{\Psi}(\xi^*) + \mathscr{K}$ can be interpreted as the set of values that are worse than or equal to $\bar{\Psi}(\xi^*)$; if ξ^* is an optimal point, then $\bar{\Psi}(\xi^*)$, the objective at ξ^*, can be compared to the objective at every other feasible point and is better than or equal to it. $\square$

Remark (On Pareto-Optimality)

Whether the vector-optimization problem $\min_\xi \bar{\Psi}(\xi)$ does not have an optimal point or optimal value, which is involved by the fact that the set of achievable objective values does not have a minimum element, then minimal elements of the set of achievable values come into play in such case.

That conducts to the notions of Pareto-optimal points and Pareto-ordering over cones, that is a feasible point ξ is Pareto-optimal or efficient if $\bar{\Psi}(\xi)$ is a minimal element of the set of achievable values Γ. $\square$

Pareto optimality III

Definition (Pareto-Optimal Points and Pareto-Optimal Values)

A point ξ is Pareto-optimal if and only if it is feasible and

$$\left(\bar{\Psi}(\xi) - \mathscr{K}\right) \cap \Gamma = \left\{\bar{\Psi}(\xi)\right\} , \tag{4}$$

that is the only achievable value better than or equal to $\bar{\Psi}(\xi)$ is $\bar{\Psi}(\xi)$ itself, whereas the set $\bar{\Psi}(\xi) - \mathscr{K}$ can be interpreted as the set of values that are better than or equal to $\bar{\Psi}(\xi)$, in this case $\bar{\Psi}(\xi)$ being a Pareto-optimal value.

A vector-optimization problem can have several Pareto-optimal points and Pareto-optimal values, respectively, and the set of Pareto-optimal values $\mathscr{P}$ satisfies

$$\mathscr{P} \subseteq \Gamma \cap \mathrm{bd}\,\Gamma ,$$

where $\mathrm{bd}\,(.)$ stands for the set boundary, that is every Pareto-optimal value is an achievable objective value that lies in the boundary of the set of achievable objective values Γ. $\square$

Scalar-inequality admissibility I

The set of admissible closed-loop models $\hat{\mathscr{A}}_f$ is defined offline, being assumed in the following that it is described using inequality constraints, precisely

$$\hat{\mathscr{A}}_f = \left\{ \hat{A}_f \mid \psi(\hat{A}_f) \leq 0 \right\} , \tag{5}$$

where $\psi : \mathbb{R}^{n \times n} \to \mathbb{R}$ is a given scalar-function, the simpler case of only one constraint being at first treated in order to draw the similarities and the differences between the admissible model matching and the classical pseudo-inverse method.

When a fault occurs, the problem becomes that of finding a control matrix K_f such that

$$\psi(A_f - B_f K_f) \leq 0 \, ; \tag{6}$$

note that, unlike the Pseudo-Inverse Method, AMM guarantees that the behavior of the accommodated system will be admissible if (6) has a solution.

Scalar-inequality admissibility II

Therefore, it is easily seen that a necessary and sufficient condition for the admissible model matching problem to have a solution is that the faulty pair (A_f, B_f) satisfies

$$\min_{K_f} \psi(A_f - B_f K_f) \leq 0\,, \tag{7}$$

where (7) involves the optimization of criterion $\psi(\hat{A}_f)$. (It follows that the set of recoverable faults is the set of pairs (A_f, B_f) such that condition (7) holds true.)

This far, (1)-(7) define the Admissible Model Matching problem.

Scalar-inequality admissibility III

In this respect, there are several good features characterizing this approach:

(i) Solving the minimization problem, for

$$K_f^* = \arg\min_{K_f} \psi(A_f - B_f K_f)\,, \tag{8}$$

is possible but it is not compulsory since it is enough to find a matrix such that (6) holds.

(ii) Finding K_f^* calls for an unconstrained optimization procedure and therefore the solution satisfies the classical stationarity condition

$$B_f^\top \frac{\partial \psi(\hat{A}_f)}{\partial \hat{A}_f} = 0\,, \tag{9}$$

assuming that ψ is differentiable.

Scalar-inequality admissibility IV

iii The approach also provides some kind of robustness; indeed, assuming that the nominal system (A_n, B_n) with nominal controller K_n runs on the time interval $[t_0, t_f)$ and that the fault (A_f, B_f) occurs at time t_f, then it is not necessary to accommodate the fault as long as

$$\psi(A_f - B_f K_n) \leq 0$$

holds, since the faulty system is driven by the nominal control in such a way that its behavior is still admissible.

Moreover, let the system (A_f, B_f) be controlled by a matrix K_f such that the closed-loop behavior is admissible; the difference

$$\Delta\psi \triangleq \psi(A_f - B_f K_f) - \psi(A_f - B_f K_f^*)$$

might be interpreted as the system robustness margin in this situation; furthermore, if K_f is the actually applied admissible control, then it is possible to improve it in order to increase the robustness margin but that can be done only at those moments when the available computation power makes it possible since this is not compulsory.

Scalar-inequality admissibility V

For example, consider the particular case when the post-fault admissible set $\hat{\mathscr{A}}_f$ is defined by

$$\psi(\hat{A}_f) = \left\| \hat{A}_f - \hat{A}_n^* \right\|_F^2 - c_{\hat{A}_f}, \tag{10}$$

where $\hat{A}_n^*$ is the ideal nominal closed-loop system and the positive scalar $c_{\hat{A}_f}$ characterizes the admissible neighborhood of the ideal system in the situation of faults impairing it; note from the theorem on the Gerschgorin disks that a choice of $c_{\hat{A}_f}$ which guarantees the system stability is

$$c_{\hat{A}_f} \leq \frac{1}{\|Z_f\|_2 \left\| Z_f^{-1} \right\|_2} \max_{\hat{\lambda}_n^*} \Lambda(\hat{A}_n^*), \text{ for any } \hat{A}_f \in \hat{\mathscr{A}}_f, \tag{11}$$

where $\Lambda(.)$ denotes the spectrum of the matrix-argument.

Scalar-inequality admissibility VI

Again, let the admissible set $\hat{\mathscr{A}}_f$ be defined by (10); hence, a necessary and sufficient condition for the fault (A_f, B_f) to be recovered is that the faulty system satisfies

$$\sum_{i=1}^{n} \left(a_f^i - \hat{a}_n^{*\,i}\right)^\top \left(I_n - B_f B_f^\dagger\right) \left(a_f^i - \hat{a}_n^{*\,i}\right) \leq c_{\hat{A}_f}\,;$$

moreover, the matrix K_f which satisfies the stationarity condition (9) is the classical solution of PIM, precisely

$$K_f = B_f^\dagger (A_f - \hat{A}_n^*)\,; \tag{12}$$

however, PIM gives only the unconstrained solution, while (12) may unfortunately saturate (10).

Furthermore, the AMM approach is very easily extended when the set $\hat{\mathscr{A}}_f$ is defined using not only one but several scalar inequalities, that is ψ in (6) is replaced by a vector function $\bar{\Psi} : \mathbb{R}^{n\times n} \to \mathbb{R}^{\bar{n}}$, with $\bar{n} > 1$.

Multi-criteria admissibility I

In any event, recall once more the consistency conditions (13), respectively

$$\mathrm{Im}\left(A_n - \hat{A}_n^*\right) \subseteq \mathrm{Im}\,(B_n)\ , \tag{13}$$

and (14), respectively

$$\mathrm{Im}\left(A_f - \hat{A}_n^*\right) \subseteq \mathrm{Im}\,(B_f)\ , \tag{14}$$

necessary for a solution to the exact model matching problem in the nominal and faulty operation situations, and introduce the definitions of feasibility and consistency of a system altogether with that of the fault recoverability as follows.

Definition (System Feasibility)

For the given pairs (A_n, B_n) and (A_f, B_f), matrix $\hat{A}_n^*$ which satisfies the consistency condition (13) or (14) is called feasible and its set is denoted by $\hat{\mathscr{A}}(A_n, B_n)$ respectively $\hat{\mathscr{A}}(A_f, B_f)$. $\square$

Multi-criteria admissibility II

Not as for the EMM, in Admissible Model Matching the problem turns to search for a control solution that will provide the closed-loop matrix such that $\hat{A}_f \in \hat{\mathscr{A}}(A_f, B_f)$ for the same ideal nominal closed-loop matrix $\hat{A}_n^*$.

Definition (System Consistency)

The post-fault system (A_f, B_f) is consistent if and only if it allows to achieve the same control objective as in the nominal case, which means that matrix $\hat{A}_n^*$ is still feasible, that is $\hat{A}_n^* \in \hat{\mathscr{A}}(A_f, B_f)$. $\square$

Regarding this, the new control matrix K_f can obviously be found by applying (15), respectively

$$K_n = B_n^{\dagger}(A_n - \hat{A}_n^*)\,, \tag{15}$$

with entries (A_f, B_f) instead of (A_n, B_n) if and only if the pair (A_f, B_f) is consistent.

Multi-criteria admissibility III

Definition (Fault Recoverability)

A fault is recoverable by a given strategy, namely system reconfiguration or fault accommodation, or it is admissible for this strategy if and only if the associated post-fault system (A_f, B_f) is consistent. $\square$

At any rate, let (A, B) stand for the pair of system and control matrices instead of (A_n, B_n) when nominal or (A_f, B_f) when faulty operation is considered, respectively; the problem consists therefore in finding a matrix K which satisfies

$$\exists\, \hat{A} \in \hat{\mathscr{A}} \text{ s.t. } A - BK = \hat{A}\,.$$

From the definitions of $\hat{\mathscr{A}}$ and $\hat{\mathscr{A}}(A, B)$, it follows that a necessary and sufficient condition for the existence of a solution is

$$\hat{\mathscr{A}}(A, B) \cap \hat{\mathscr{A}} \neq \emptyset\,. \tag{16}$$

Hence, for any consistent system (A, B), there exists a set of solutions to the admissible control problem given by

$$\mathscr{K}(A, B) = \left\{ K \mid K = B^{\dagger}(A - \hat{A})\,,\ \hat{A} \in \hat{\mathscr{A}}(A, B) \cap \hat{\mathscr{A}} \right\}\,;$$

Multi-criteria admissibility IV

furthermore, the recoverable faults are such that the pair (A, B) is consistent, which means that the set of solutions $\mathscr{K}(A, B)$ is not empty; the definition of a consistent system has therefore to be seen in the sense that (A, B) is consistent if and only if it satisfies (16).

In practice, it is enough to find a matrix $\hat{A}$ such that

$$\bar{\Psi}(\hat{A}) \leq 0 \qquad \text{and} \tag{17a}$$

$$(I_n - BB^\dagger)(A - \hat{A}) = 0\,, \tag{17b}$$

since the second condition ensures that it is feasible.

By rewriting condition (17b) under the form

$$J(\hat{A}) = \sum_{i=1}^{n} \left(a^i - \hat{a}^i\right)^\top \left(I_n - BB^\dagger\right) \left(a^i - \hat{a}^i\right) = 0\,,$$

with the assumption that $I_n - BB^\dagger \geq 0$, dual solutions are suggested as follows.

Multi-criteria admissibility V

The first approach turns to solving the constrained optimization problem

$$\min_{\hat{A}\in\hat{\mathscr{A}}} J(\hat{A}) \text{ s.t. } \bar{\Psi}(\hat{A}) \leq 0\,,$$

that is a classical quadratic optimization problem under inequality constraints; in this case, a necessary and sufficient condition for the admissible model matching problem to have a solution for the system (A, B) is that $J(\hat{A}^*) = 0$ if $\hat{A}^*$ is a solution; therefore, when a solution exists, it will be found whatever the norm used. (See that there is no drawback inhere using the Frobenius-norm in the problem definition; indeed, the norm is not interpreted as the distance between two systems as in PIM but it is used only to find, when it exists, an admissible matrix $\hat{A} \in \hat{\mathscr{A}}$ which is also feasible, i.e., $\hat{A} \in \hat{\mathscr{A}}(A, B)$.)

Vector criterion optimization and Pareto optimal solutions I

The second approach turns however to solving the multi-criteria optimization problem

$$\min_{\hat{A}\in\hat{\mathscr{A}}} \bar{\Psi}(\hat{A}) \text{ s.t. } J(\hat{A}) = 0\,, \tag{18}$$

that is a Multi-Objective Program (MOP); in this case, (18) can be rewritten as an unconstrained but multi-criteria optimization problem after substituting for the constraint, precisely

$$\min_{K} \bar{\Psi}(A - BK)\,. \tag{19}$$

Hence, let $\mathscr{P}(A, B)$ be the set of Pareto-optimal solutions of (19), i.e., the set of matrices K^* such that (20) is true, respectively

$$\forall\, K \notin \mathscr{P}(A, B)\,,\ \exists\, K^* \in \mathscr{P}(A, B) \text{ s.t. } K^* \succ K\,, \tag{20a}$$

$$\forall\, K_1^* \in \mathscr{P}(A, B) \text{ and } \forall\, K_2^* \in \mathscr{P}(A, B)\,,\ K_1^* \succeq K_2^* \Rightarrow K_2^* \succeq K_1^*\,, \tag{20b}$$

where (20a) means that, whatever the non-Pareto-optimal matrix K, there is a Pareto-optimal one K^* that strictly dominates it, being better at least for one and worse for none of the criteria, while (20b) means that Pareto-optimal solutions are not comparable (should K_1^* dominate K_2^* then K_2^* dominates K_1^* and therefore they are equivalent, because no Pareto-optimal solution can be preferred to another Pareto-optimal one).

Vector criterion optimization and Pareto optimal solutions II

Therefore, a necessary and sufficient condition for the admissible model matching problem to have a solution for the system (A, B) is that

$$\exists\, K \in \mathscr{P}(A, B) \text{ s.t. } \bar{\Psi}(A - BK) \leq 0\,; \tag{21a}$$

precisely, condition (21a) extends to

$$\mathscr{P}_K\left[\bar{\Psi}(A - BK)\right] \cap \mathscr{K}^- \neq \emptyset\,, \tag{21b}$$

meaning that the set of Pareto-optimal solutions intersects the negative-cone in the $\mathscr{K}$-space.

Pareto optimal admissible control and passive robustness I

Furthermore, assume that, due to the occurrence of a fault, the pair (A_n, B_n) is changed into (A_f, B_f) and that $K_n \in \mathscr{K}(A_f, B_f)$; then, the closed-loop control K_n is still acceptable and nothing has to be done in response to the fault; moreover, given a feedback K_n, the set of all pairs (A_f, B_f) such that $K_n \in \mathscr{K}(A_f, B_f)$ is the set of all systems for which the closed-loop control K_n is admissible.

Hence, let

$$\hat{\mathscr{A}}_f(K_n) = \left\{ (A_f, B_f) \mid A_f - B_f K_n \in \hat{\mathscr{A}}_f \right\}$$

be that set; it follows that, when changes that are the result of faults occur in the pair (A_n, B_n), passive fault tolerant control is achieved by the feedback control matrix K_n as long as the system (A_f, B_f) remains in $\hat{\mathscr{A}}_f(K_n)$.

Therefore, it may be interesting to evaluate the passive fault tolerance of system (A_n, B_n) under the control K_n. Let $M_n \triangleq (A_n, B_n)$, $M_f \triangleq (A_f, B_f)$ and $d(M_n, M_f)$ be some distance defined between the two systems given by their models M_n and M_f, respectively by their system and control matrices pairs (A_n, B_n) and (A_f, B_f).

Pareto optimal admissible control and passive robustness II

A natural robustness measure is provided by the two indexes $\delta_m(M_n, K_n)$ and $\delta_M(M_f, K_n)$, precisely

$$\delta_m(M_n, K_n) = \inf_{M_f \notin \hat{\mathscr{A}}_f(K_n)} d(M_n, M_f)\,,$$
$$\delta_M(M_f, K_n) = \sup_{M_f \in \hat{\mathscr{A}}_f(K_n)} d(M_n, M_f)\,,$$

whereas $\delta_m(M_n, K_n)$ is the radius of a ball in the systems-space centered on system M_n such that the control K_n is admissible for any system within that ball, while $\delta_M(M_f, K_n)$ is the radius of a ball in the systems-space centered on system M_f such that the control K_n is not admissible whatever the system outside that ball.

A simple result on the 2-D case, the ball becoming a circle, for the B747 short-period mode is presented in the following.

Outline

1. Admissible Model Matching
2. B747 Application Example
   - Admissible Model Matching

Admissible Model Matching I

The set of admissible closed-loop matrices is first specified in the admissible model matching (AMM), so as to account for closed-loop system performance.

Let M be a closed-loop matrix with eigenvalues $\Lambda(M) = \{\lambda_1, \lambda_2\}$.

The undamped natural frequency and the damping coefficient are $\omega = \sqrt{\lambda_1 \lambda_2}$ and $\zeta = -\frac{\lambda_1 + \lambda_2}{\sqrt{\lambda_1 \lambda_2}}$.

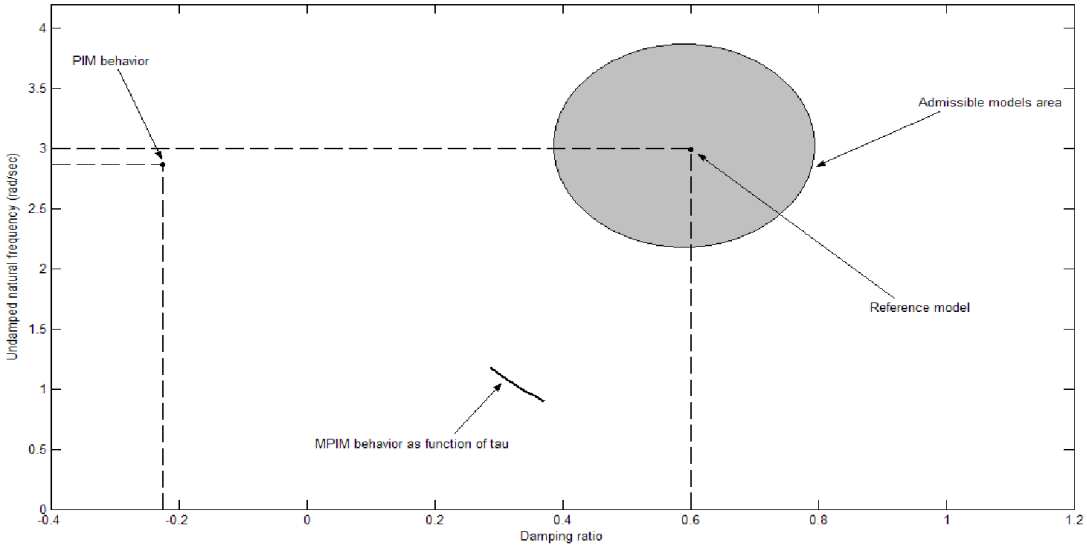
According to the "thumb-print" criterion, all closed-loop matrices with satisfactory short-period mode performance are described by

$$\Phi_1(M) = \frac{(\zeta - 0.6)^2}{0.04} + \frac{(\omega - 3)^2}{0.64} - 1 \leq 0, \tag{22}$$

which is taken as the first condition for admissibility.

Admissible Model Matching II

The following figure depicts the set of acceptable pairs (ω, ζ).



Admissible models set

Moreover, it shows the unstable one provided by the PIM, and also the stable but unsatisfactory ones provided by the MPIM.

Admissible Model Matching III

Note that defining admissibility via relation (22) is not very demanding.

Indeed, if the pair (A_f, B_f) is still controllable, there is always a feedback law K_f such that $\Lambda(M) = \Lambda(M^*)$, and therefore $\Phi_1(M) = -1$ is obtained.

It follows that specifying admissibility only by (22) boils down to designing fault accommodation on the basis of eigenvalues assignment, and in that case, any fault such that the system is still controllable can be recovered.

Note that, under this strategy, the discrepancies in the system trajectories can be easily computed by

$$e(t) = V^* \left[T e^{(\Lambda^* t)} T^{-1} - e^{(\Lambda^* t)} \right] (V^*)^{-1} x_0, \tag{23}$$

where $e(t) = x(t) - x^*(t)$, $x(t)$ is the solution of the accommodated closed-loop, $x^*(t)$ is the solution of the reference closed-loop, x_0 is the initial condition, $\Lambda^* = diag\left(\lambda_1^*, \lambda_2^*\right) = (V^*)^{-1} M^* V^* = (V)^{-1} MV$, and $T = (V^*)^{-1} V$.

Admissible Model Matching IV

From (23), one has

$$\|e(t)\| \leq \eta \left(1 + \|T\| \left\|T^{-1}\right\|\right) \left\|e^{(\Lambda^* t)}\right\| \|x_0\|, \tag{24}$$

where $\eta = \|V^*\| \left\|(V^*)^{-1}\right\|$.

It follows that a second condition to define admissible closed-loops is naturally obtained, by imposing a given bound on the discrepancy by means of

$$\Phi_2(M) = \|T\| \left\|T^{-1}\right\| - \alpha \leq 0, \tag{25}$$

where α is a given constant.

Note that defining the set of admissible closed-loops by $\Phi_1(M) \leq 0$ and $\Phi_2(M) \leq 0$ establishes a clear link between recoverable faults and acceptable performances.

Admissible Model Matching V

Let, for example, the bound on the distance between the system trajectories be defined by $\alpha = 1.10$.

According to the admissible model matching development, the fault defined by the pair $(\gamma, \tau) = (0.05, 0.5)$ will give the system and control matrices $A_f = 0.95 A_n$ and $B_f = 0.5 B_n$, respectively

$$A_f = \begin{bmatrix} -0.2971 & 735.3000 \\ -0.0010 & -0.4070 \end{bmatrix}, \; B_f = \begin{bmatrix} -8.9708 \\ -0.5788 \end{bmatrix}. \tag{26}$$

Admissible Model Matching VI

The control matrix

$$K_{AMM_1} = \begin{bmatrix} -0.0173 & -4.7356 \end{bmatrix} \tag{27}$$

achieves the closed-loop behavior

$$M_{AMM_1} = \begin{bmatrix} -0.4518 & 692.8183 \\ -0.0109 & -3.1482 \end{bmatrix}, \tag{28}$$

whose poles

$$\Lambda\left(M_{AMM_1}\right) = \{-1.8 \pm 2.4\jmath\} \tag{29}$$

are obviously the reference ones, so that the dynamic performance parameters $(\omega^{AMM_1} = 3.0, \zeta^{AMM_1} = 0.6)$ satisfy the two conditions $\Phi_1(M_{AMM_1}) = -1 \leq 0$ and $\Phi_2(M_{AMM_1}) = -0.04 \leq 0$.

Therefore, the fault $(\gamma = 0.05, \tau = 0.5)$ is recoverable.

Admissible Model Matching VII

On the contrary, the fault defined by the pair $(\gamma, \tau) = (0.1, 0.5)$ will give the system and control matrices $A_f = 0.9A_n$ and $B_f = 0.5B_n$, respectively

$$A_f = \begin{bmatrix} -0.2814 & 696.6000 \\ -0.0009 & -0.3856 \end{bmatrix}, \; B_f = \begin{bmatrix} -8.9708 \\ -0.5788 \end{bmatrix}. \tag{30}$$

Admissible Model Matching VIII

In this case, the control matrix

$$K_{AMM_2} = \begin{bmatrix} -0.0185 & -4.7802 \end{bmatrix} \tag{31}$$

achieves the closed-loop behavior

$$M_{AMM_2} = \begin{bmatrix} -0.4474 & 653.7175 \\ -0.0116 & -3.1526 \end{bmatrix} \tag{32}$$

with poles

$$\Lambda\left(M_{AMM_2}\right) = \Lambda\left(M_{AMM_1}\right). \tag{33}$$

However, even the dynamic parameters pair $(\omega^{AMM_2} = \omega^{AMM_1}, \zeta^{AMM_2} = \zeta^{AMM_1})$ stands for the first condition, $\Phi_1(M_{AMM_2}) = -1 \leq 0$, it does not hold for the second one, since $\Phi_2(M_{AMM_2}) = 0.04 \not\leq 0$, and therefore the fault $(\gamma, \tau) = (0.1, 0.5)$ is a non-recoverable one.

Questions? vs. Answers!

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